

2.

$$E_x = E_0 \sin(\beta_y y) e^{-jz} \Rightarrow \beta_x = 0, \beta_y = \beta_y, \beta_z = -j\delta$$

$$(a) \beta_x^2 + \beta_y^2 + \beta_z^2 = \beta^2 \Rightarrow \beta_y^2 - \delta^2 = \beta^2 \Rightarrow \delta^2 = \beta_y^2 - \beta^2$$

$$\delta = \sqrt{\beta_y^2 - \beta^2}$$

$$\nabla^2 E_x + \beta^2 E_x = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) E_x + \beta^2 E_x = -\beta_y^2 + \delta^2 + \beta^2 = 0$$

$\therefore H^0$ is defining a set of TE modes.

$$(b) E_x(y=0) = E_0(0) e^{-jz} = 0$$

$$E_x(y=b) = E_0 \sin(\beta_y b) e^{-jz} = 0 \Rightarrow \sin(\beta_y b) = 0$$

$$\beta_y b = n\pi; n = 1, 2, 3, \dots$$

$$\beta_y = \left(\frac{n\pi}{b} \right), n = 1, 2, 3, 4, \dots$$

$$\delta = \sqrt{\beta_y^2 - \beta^2} = \sqrt{\left(\frac{n\pi}{b} \right)^2 - \beta^2} = j \sqrt{\beta^2 - \left(\frac{n\pi}{b} \right)^2} \text{ for } f > f_c$$

$$\beta_c = \omega \sqrt{\mu \epsilon} = \beta_y = \frac{n\pi}{b} \Rightarrow \omega_c = \frac{1}{\sqrt{\mu \epsilon}} \left(\frac{n\pi}{b} \right) = 2\pi f_c$$

$$f_c = \frac{n}{2b \sqrt{\mu \epsilon}}, n = 1, 2, 3, \dots$$

$$H_z = \frac{1}{j\omega\mu} \nabla \times E = \frac{-1}{j\omega\mu} \left[\hat{a}_x(0) \times \hat{a}_y \left(\frac{\delta E_x}{\delta z} \right) - \hat{a}_z \frac{\delta E_x}{\delta y} \right]$$

$$H_y = \frac{-1}{j\omega\mu} \frac{\delta E_x}{\delta z} = \frac{\delta}{j\omega\mu} E_0 \sin(\beta_y y) e^{-jz} \Big|_{\delta = j\beta_z} = \frac{\beta_z}{\omega\mu} E_0 \sin(\beta_y y) e^{j\beta_z z}$$

$$P = \frac{1}{2} \iint_{\partial V} \text{Re}(E \times H^*) \cdot dS = \frac{1}{2} \iint \text{Re}[\hat{a}_x E_x \times (\hat{a}_y H_y^* + \hat{a}_z H_z^*)] \cdot \hat{a}_z dy dz$$

$$= \frac{1}{2} \iint_{\partial V} \text{Re}(\hat{a}_z E_x H_y^* - \hat{a}_y E_x H_z^*) \cdot \hat{a}_z dy dz = \text{Re} \left\{ \frac{1}{2} \int_0^b \int_0^L (E_x H_y^* dy dz) \right\}$$

$$= \text{Re} \left\{ \frac{1}{2} \int_0^b \int_0^L |E_0|^2 \sin^2(\beta_y y) \frac{\beta_z}{\omega\mu} dy dz \right\} = \frac{|E_0|^2 \beta_z}{4 \omega\mu} \int_0^b (1 - \cos(2\beta_y y)) dy$$

$$P = \frac{|E_0|^2 \beta_z}{4 \omega\mu} \left[y - \frac{1}{2\beta_y} \sin(2\beta_y y) \right]_0^b = \frac{|E_0|^2 b \beta_z}{4 \omega\mu}$$

$$\boxed{\frac{P}{2} = \frac{|E_0|^2 b \beta_z}{4 \omega\mu} \sqrt{1 - \left(\frac{f_c}{f} \right)^2}}$$

③ a) Given $n = 2^N \frac{\eta_1 - \eta_0}{\eta_1 + \eta_0} c^N$

$$\eta_1 = 258.56 \Omega$$

$$\eta_0 = 499.5 \Omega$$

$$n=0: \Gamma_0 = 2^2 \frac{258.56 - 499.5}{258.56 + 499.5} \times \frac{21}{2101} = \frac{1}{4} (-0.318) = -0.0795$$

$$\Rightarrow \frac{\eta_1 - \eta_0}{\eta_1 + \eta_0} = -0.0795$$

$$\eta_1 = \eta_0 \left(\frac{1 - 0.0795}{1 + 0.0795} \right)$$

$$\boxed{\eta_1 = 425.928 \Omega}$$

$$n=1: \Gamma_1 = 2^2 \frac{258.56 - 499.5}{258.56 + 499.5} \times \frac{21}{1111} = -0.159 = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\boxed{\eta_2 = 309.03 \Omega}$$

b) $\eta_1 = \frac{377/\sqrt{\epsilon_{r1}}}{\sqrt{1 - \left(\frac{6.56}{10\sqrt{\epsilon_{r1}}}\right)^2}} \Rightarrow \epsilon_{r1} = 0.43 + \left(\frac{377}{\eta_1}\right)^2 = 1.213$

$$\boxed{\epsilon_{r1} = 1.213}$$

$$\eta_2 = \frac{377/\sqrt{\epsilon_{r2}}}{\sqrt{1 - \left(\frac{6.56}{10\sqrt{\epsilon_{r2}}}\right)^2}} \Rightarrow \epsilon_{r2} = 0.43 + \left(\frac{377}{\eta_2}\right)^2$$

$$\boxed{\epsilon_{r2} = 1.918}$$

c) $L_1 = \frac{\lambda_{g1}}{4} = \frac{\eta_0/\sqrt{\epsilon_{r1}}}{4\sqrt{1 - \left(\frac{6.56}{10\sqrt{\epsilon_{r1}}}\right)^2}} = \frac{3/\sqrt{1.213}}{4\sqrt{1 - \frac{0.43}{\epsilon_{r1}}}}$

$$\boxed{L_1 = 0.8476 \text{ cm}}$$

$$L_2 = \frac{\lambda_{g2}}{4} = \frac{\eta_0/\sqrt{\epsilon_{r2}}}{4\sqrt{1 - \left(\frac{6.56}{10\sqrt{\epsilon_{r2}}}\right)^2}} = \frac{3/\sqrt{1.918}}{4\sqrt{1 - \frac{0.43}{\epsilon_{r2}}}}$$

$$\boxed{L_2 = 0.6148 \text{ cm}}$$

$$4. (a) TE^z: F_z(x, y, z) = [C_1 \cos(\beta_x x) + D_1 \sin(\beta_x x)] [C_2 \cos(\beta_y y) + D_2 \sin(\beta_y y)] A_3 e^{-j\beta_z z}$$

$$\beta_x^2 + \beta_y^2 + \beta_z^2 = \beta^2$$

$$E_x = -\frac{1}{\epsilon} \frac{\partial F_z}{\partial y}$$

$$E_y = -\frac{1}{\epsilon} \frac{\partial F_z}{\partial x}$$

$$F_z = -\frac{1}{\epsilon} \frac{\partial F_z}{\partial z} = 0$$

$$H_x = -\frac{j}{\omega \mu \epsilon} \frac{\partial^2 F_z}{\partial y \partial z}$$

$$H_y = -\frac{j}{\omega \mu \epsilon} \frac{\partial^2 F_z}{\partial x \partial z}$$

$$H_z = \frac{j}{\omega \mu \epsilon} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) F_z$$

B.C.s

$$E_x(0 \leq x \leq a, y=0, z) = 0 = E_x(0 \leq x \leq a, y=b, z)$$

$$H_y(x=0, 0 \leq y \leq b, z) = 0 = H_y(x=a, 0 \leq y \leq b, z)$$

$$E_x = -\frac{1}{\epsilon} \frac{\partial F_z}{\partial y} = -\frac{\beta_y}{\epsilon} [C_1 \cos(\beta_x x) + D_1 \sin(\beta_x x)] [-C_2 \sin(\beta_y y) + D_2 \cos(\beta_y y)] A_3 e^{-j\beta_z z}$$

$$E_x(y=0) = -\beta_y [C_1 \cos(\beta_x x) + D_1 \sin(\beta_x x)] [-C_2(1) + D_2(0)] A_3 e^{-j\beta_z z} = 0$$

$$\boxed{D_2 = 0}$$

$$E_x(x=a) = 0 \Rightarrow \sin \beta_x a = 0$$

$$\boxed{\beta_x = \frac{n\pi}{a}}$$

$$n = 0, 1, 2, 3, \dots$$

$$F_z(x, y, z) = C_2 [C_1 \cos(\beta_x x) + D_1 \sin(\beta_x x)] \cos(\beta_y y) A_3 e^{-j\beta_z z}$$

$$H_y = -\frac{j}{\omega \mu \epsilon} \frac{\partial^2 F_z}{\partial x \partial z} = +\frac{j \beta_x (-j\beta_z)}{\omega \mu \epsilon} C_2 [C_1 \cos(\beta_x x) + D_1 \sin(\beta_x x)] \cos(\beta_y y) A_3 e^{-j\beta_z z}$$

$$B.C.: H_y(x=0) = \frac{j \beta_x (-j\beta_z)}{\omega \mu \epsilon} C_2 [C_1(1) + D_1(0)] \sin(\beta_y y) A_3 e^{-j\beta_z z} = 0$$

$$\boxed{C_1 = 0}$$

$$H_y(x=a) = \frac{j \beta_x (-j\beta_z)}{\omega \mu \epsilon} C_2 [D_1 \sin(\beta_x a)] \sin(\beta_y y) A_3 e^{-j\beta_z z} = 0$$

$$\sin(\beta_x a) = 0 \Rightarrow \beta_x a = 0 = \sin^{-1}(0) = m\pi$$

$$\beta_x = \frac{m\pi}{a}, \quad m = 1, 2, \dots$$

$$F_z(x, y, z) = A_{mn} \sin(\beta_x x) \cos(\beta_y y) e^{-j\beta_z z} \quad \left. \begin{array}{l} \beta_x = \frac{m\pi}{a} \quad m=1, 2, \dots \\ \beta_y = \frac{n\pi}{b} \quad n=0, 1, 2, \dots \end{array} \right\}$$

(b) Lossless Mode

$$(f_c)_{TE} = \frac{1}{2a\sqrt{\mu\epsilon}} \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

$$(f_c)_{10} = \frac{1}{2a\sqrt{\mu_0\epsilon_0\epsilon_r}} \left(\frac{\pi}{a}\right) = \frac{1}{2a\sqrt{\mu_0\epsilon_0\epsilon_r}} = \frac{30 \times 10^9}{2(2.286)(1)} = 6.58 \times 10^9 \text{ Hz}$$

$$(f_c)_{10} = \frac{30 \times 10^9}{2(2.286)} = 6.58 \text{ GHz}$$

(5) (a) Since there are no x or y variations, then the electric field must be represented as a standing wave
Thus,

$$E_y = [C_1 \cos(\beta_z z) + D_1 \sin \beta_z z]$$

$$E_y(z=0) = 0 \Rightarrow C_1(1) + D_1(0) = 0 \Rightarrow C_1 = 0$$

$$E_y(z=d) = 0 \Rightarrow D_1 \sin \beta_z d = 0$$

$$\beta_z = \frac{n\pi}{d}, n = 1, 2, 3, \dots$$

Thus, $E_y = D_1 \sin(\beta_z z) = E_0 \sin(\beta_z z)$

$$\beta_x^2 + \beta_y^2 + \beta_z^2 = \beta^2 = \omega^2 \mu \epsilon \Rightarrow \beta_z = \beta = \omega \sqrt{\mu \epsilon}$$

$$E_y = E_0 \sin(\beta z) \Rightarrow H_x = \frac{1}{j\omega \mu} \nabla \times E = -\frac{1}{j\omega \mu} \left[\hat{a}_z \frac{\partial E_y}{\partial z} \right]$$

$$= -j \frac{\beta E_0}{\omega \mu} \hat{a}_z \cos(\beta z)$$

$$H_x = -j \frac{\omega \sqrt{\mu \epsilon} E_0}{\omega \mu} \cos(\beta z) = -j \sqrt{\frac{\epsilon}{\mu}} \cos(\beta z)$$

(b) $\beta_z = \frac{n\pi}{d} \Rightarrow d = \frac{n\pi}{\beta_z} = \frac{n\pi}{\beta} = \frac{n\pi}{\frac{\omega}{c}} = \frac{n\pi c}{\omega}$ for $n = 1, 2, 3, \dots$

(c) $\omega = 2\omega_c = 2 \left[\frac{c_0}{4} \int \int \int_V |E|^2 dV \right] = \frac{c_0}{2} \int_0^a \int_0^b \int_0^d |E_0|^2 \sin^2 \beta z dz dy dx$

$$= \frac{abdc_0}{4} |E_0|^2$$

$$P_d = 2 \left\{ \frac{R_m}{2} \right\} \int_0^{\eta} d\eta = R_m \int_0^{\eta} \frac{|E_0|^2}{\eta^2} d\eta$$

$$Q = \frac{W}{P_d} = \frac{W}{R_m \int_0^{\eta} \frac{|E_0|^2}{\eta^2} d\eta} = \frac{W \epsilon_0 d}{4 R_m}$$

$$(2) R_m = \sqrt{\frac{W \epsilon_0 d}{2 \delta}} = \sqrt{\frac{2\pi (60 \times 10^9) 4\pi \times 10^{-7}}{2(5.76 \times 10^{-3})}} = 6.4127 \times 10^{-2}$$

$$d = 5\lambda = 5 \left(\frac{2\pi \times 10^9}{60 \times 10^9} \right) = 2.5 \text{ cm} = 2.5 \times 10^{-2} \text{ m}$$

$$Q = \frac{W \epsilon_0 d}{4 R_m} = \frac{2\pi \times 60 \times 10^9 (10^9 / 36\pi) (2.5 \times 10^{-2}) (377)}{4 (6.4127 \times 10^{-2})}$$

$$= 46,237.296$$

$$Q \approx 46,237$$

$$d = 10\lambda : Q = 2(46237.296) = 92474.592$$

$$Q \approx 92,475$$